

Analyse et Prédiction des Maladies Cardiaques

Statistiques des modèles - Ensemble de test

Version corrigée - Avril 2025

Performances des modèles - Ensemble de test

Modèle	Accuracy	Precision	Recall	F1-Score	AUC	TP	TN	FP	FN
Logistic Regression	81.3%	79.2%	86.2%	82.5%	89.9%	175	146	46	28
Decision Tree	80.0%	77.2%	86.7%	81.7%	85.3%	176	140	52	27
KNN	84.1%	81.2%	89.7%	85.2%	88.9%	182	150	42	21
Random Forest	83.5%	79.7%	91.1%	85.1%	92.3%	185	145	47	18
AdaBoost	81.5%	80.1%	85.2%	82.6%	89.7%	173	149	43	30
SVM	83.5%	82.9%	85.7%	84.3%	90.6%	174	156	36	29
Precision Ensemble	83.8%	79.8%	91.6%	85.3%	91.5%	186	145	47	17

Détails - Logistic Regression

Métriques principales

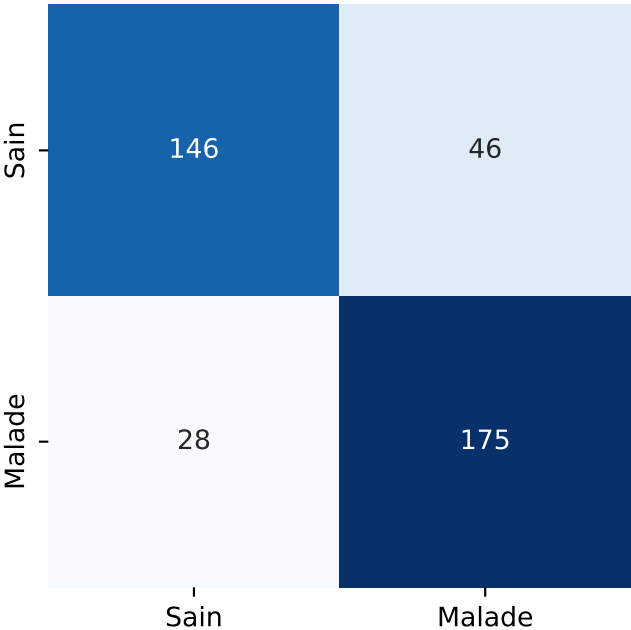
Accuracy: 81.3%

Precision: 79.2%

Recall: 86.2%

F1-Score: 82.5%

Matrice de confusion



Détail matrice

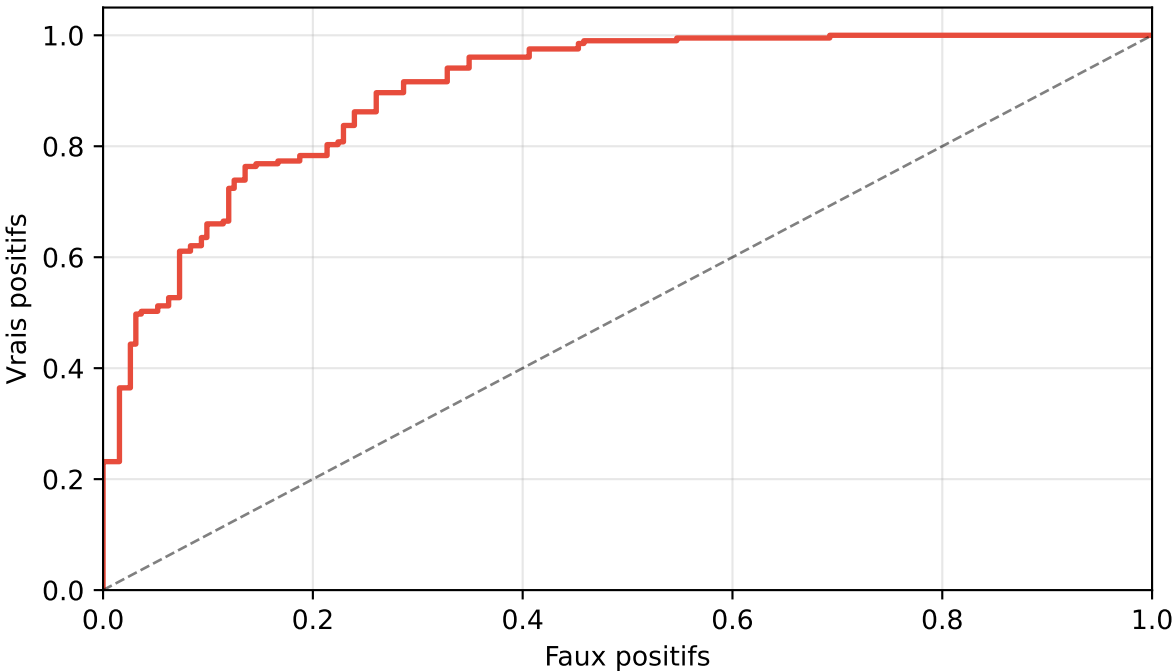
TP: 175

TN: 146

FP: 46

FN: 28

ROC (AUC = 0.899)



Détails - Decision Tree

Métriques principales

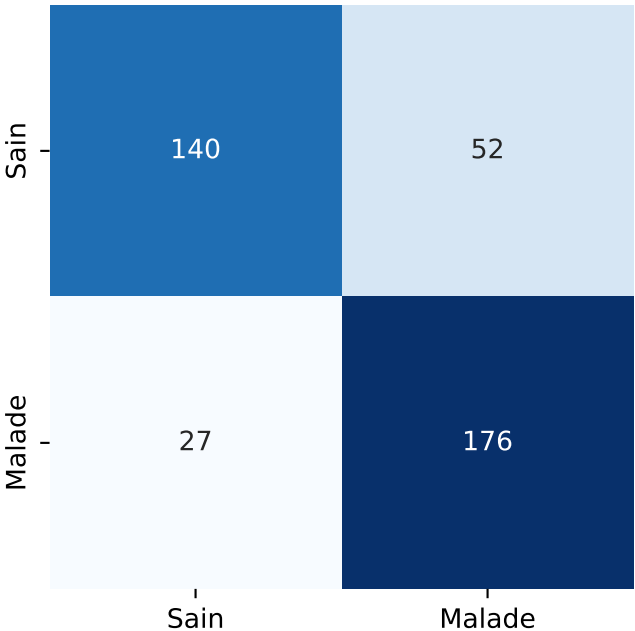
Accuracy: 80.0%

Precision: 77.2%

Recall: 86.7%

F1-Score: 81.7%

Matrice de confusion



Détail matrice

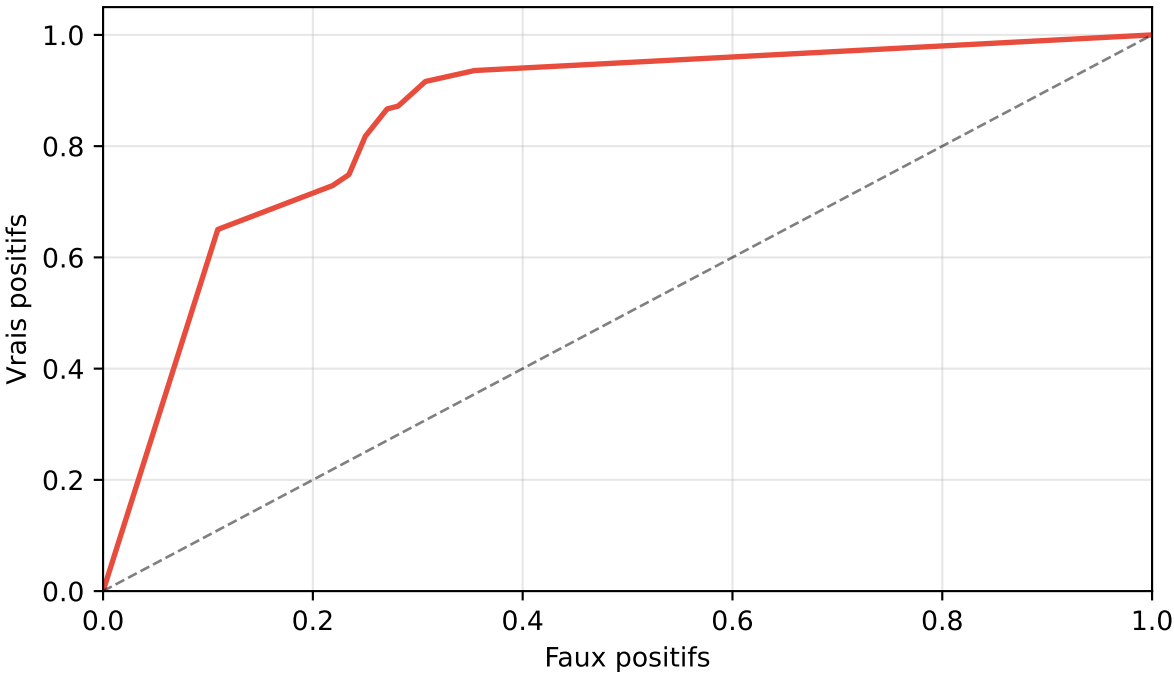
TP: 176

TN: 140

FP: 52

FN: 27

ROC (AUC = 0.853)



Détails - KNN

Métriques principales

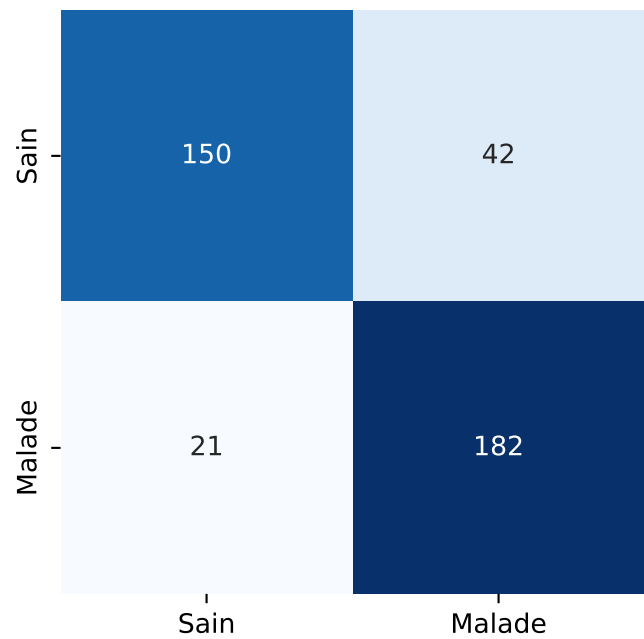
Accuracy: 84.1%

Precision: 81.2%

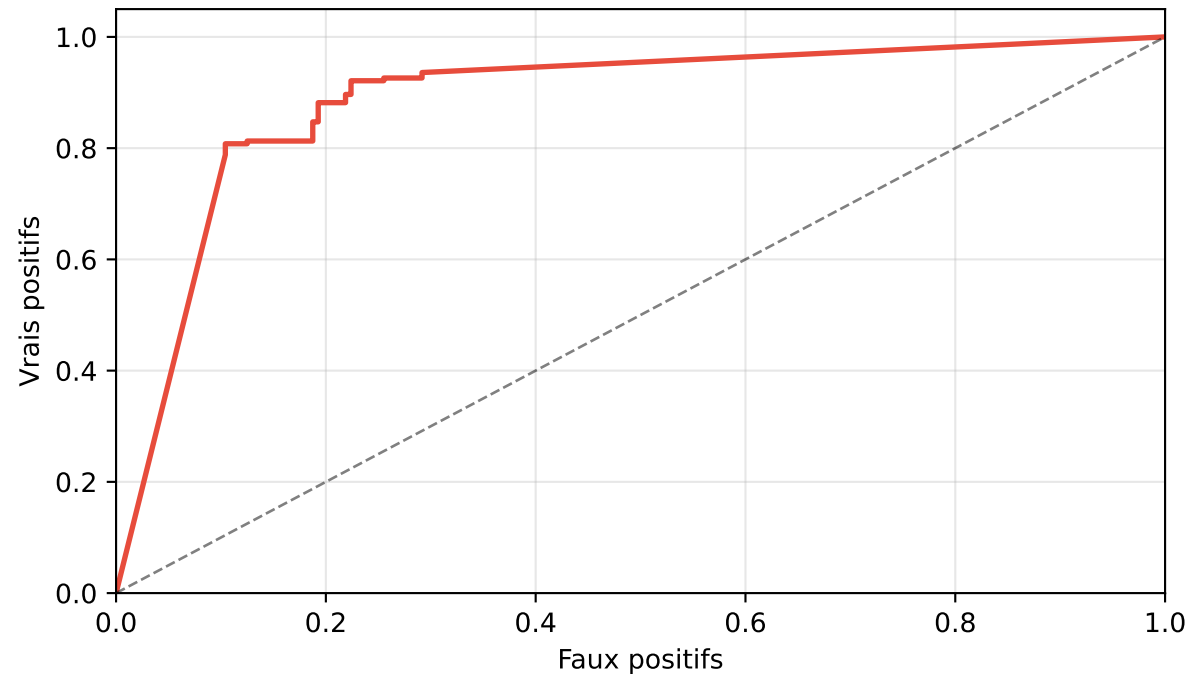
Recall: 89.7%

F1-Score: 85.2%

Matrice de confusion



ROC (AUC = 0.889)



Détail matrice

TP: 182

TN: 150

FP: 42

FN: 21

Détails - Random Forest

Métriques principales

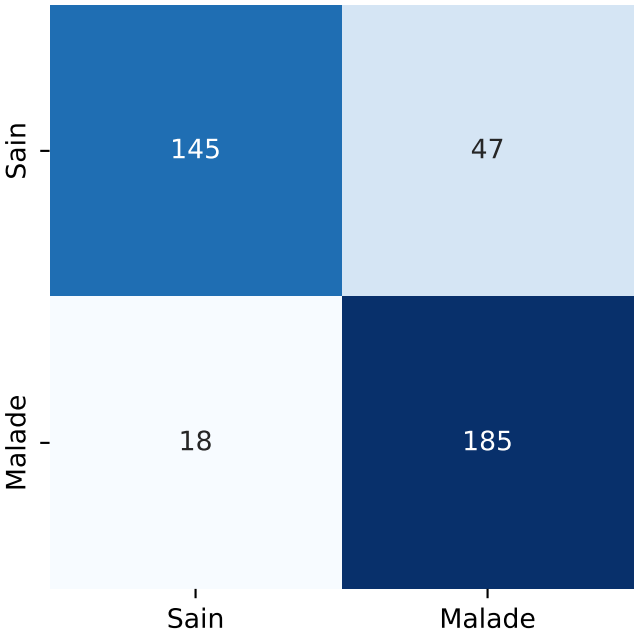
Accuracy: 83.5%

Precision: 79.7%

Recall: 91.1%

F1-Score: 85.1%

Matrice de confusion



Détail matrice

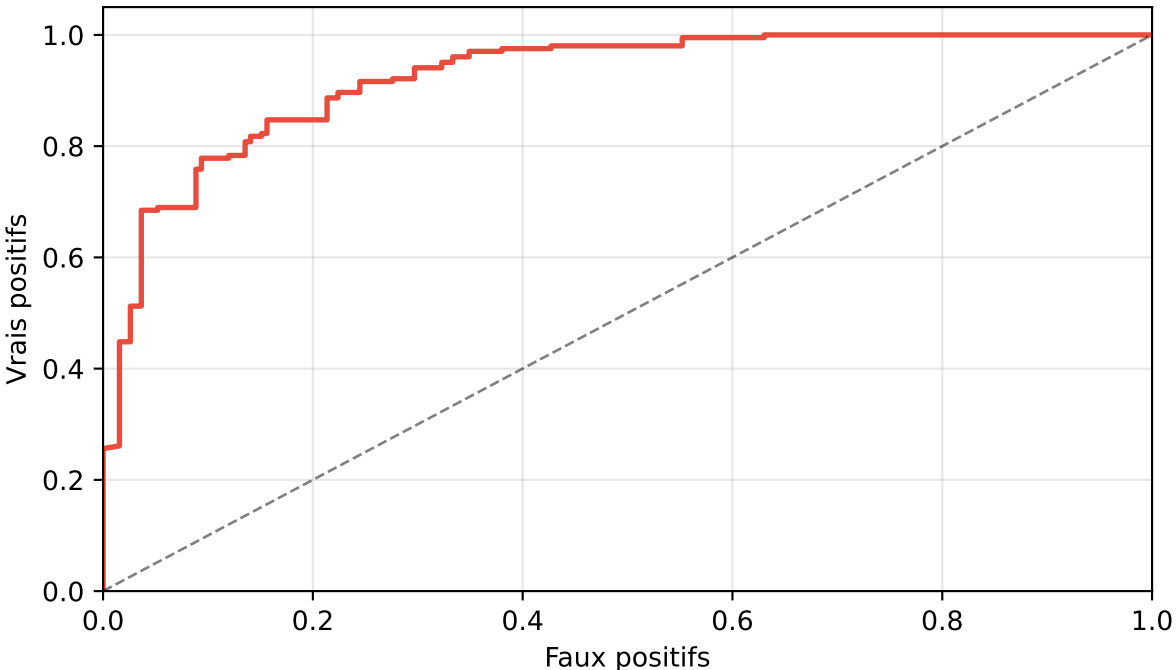
TP: 185

TN: 145

FP: 47

FN: 18

ROC (AUC = 0.923)



Détails - AdaBoost

Métriques principales

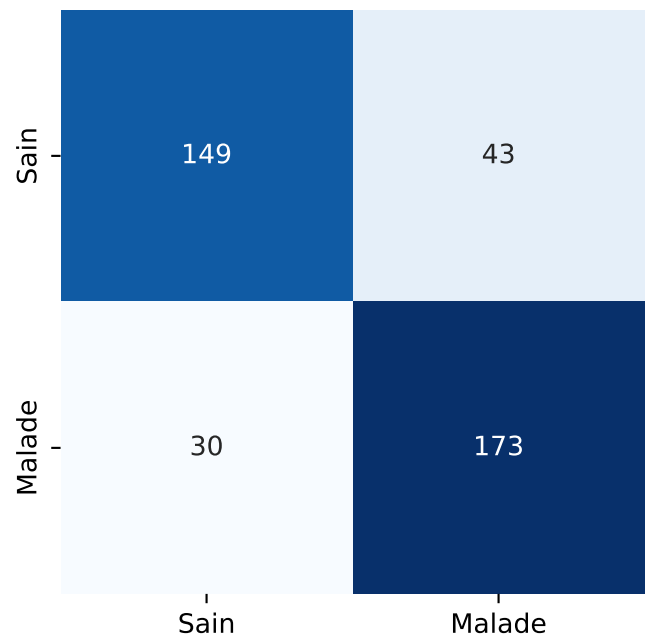
Accuracy: 81.5%

Precision: 80.1%

Recall: 85.2%

F1-Score: 82.6%

Matrice de confusion



Détail matrice

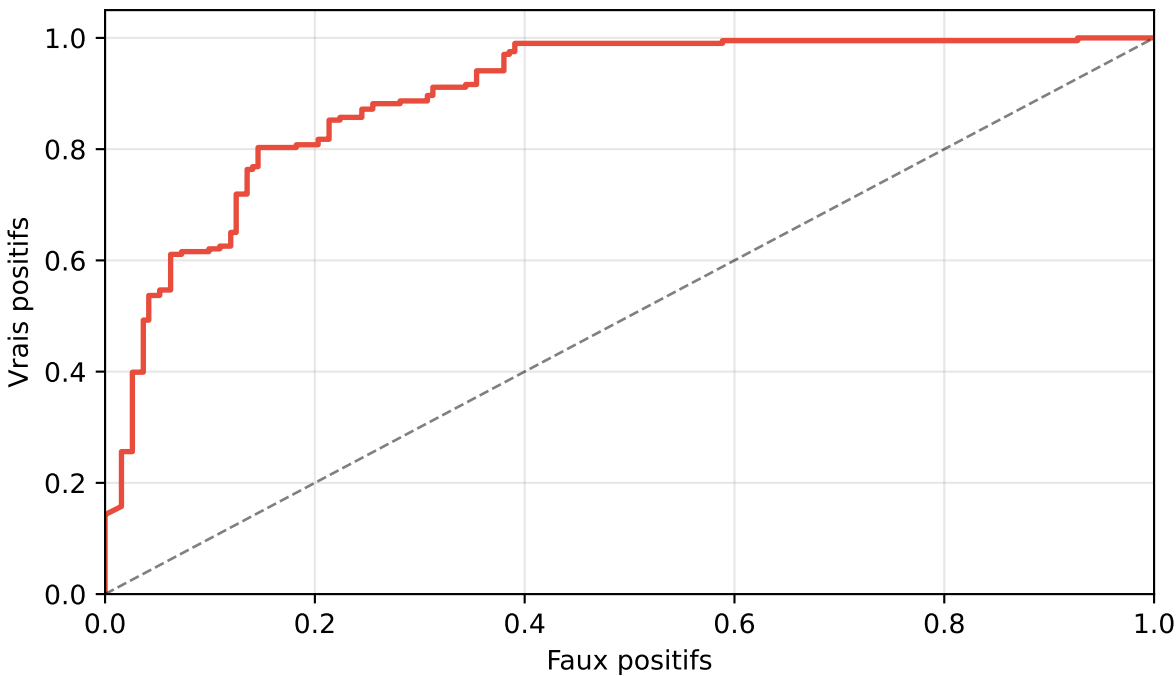
TP: 173

TN: 149

FP: 43

FN: 30

ROC (AUC = 0.897)



Détails - SVM

Métriques principales

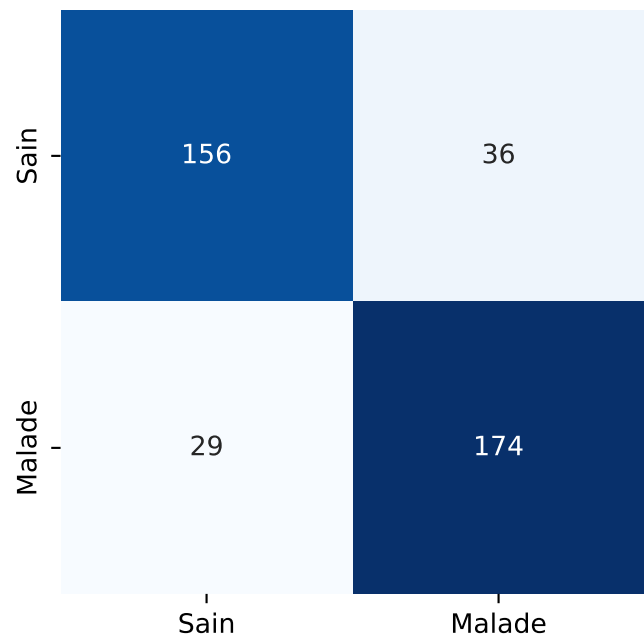
Accuracy: 83.5%

Precision: 82.9%

Recall: 85.7%

F1-Score: 84.3%

Matrice de confusion



Détail matrice

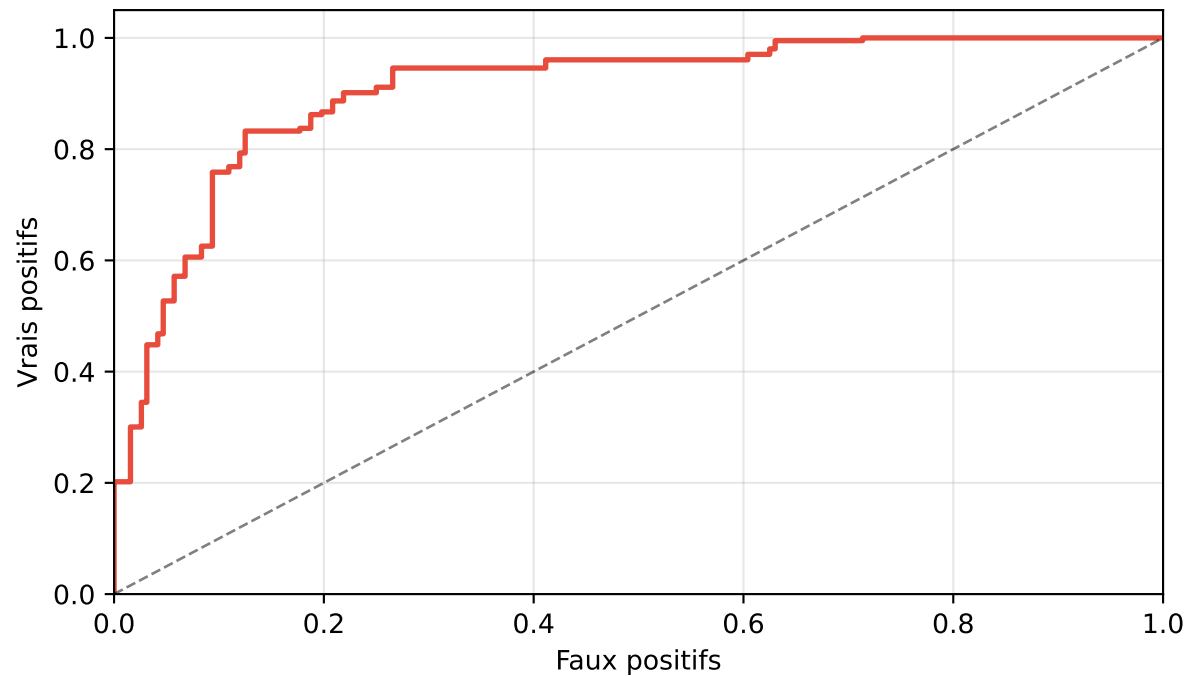
TP: 174

TN: 156

FP: 36

FN: 29

ROC (AUC = 0.906)



Détails - Precision Ensemble

Métriques principales

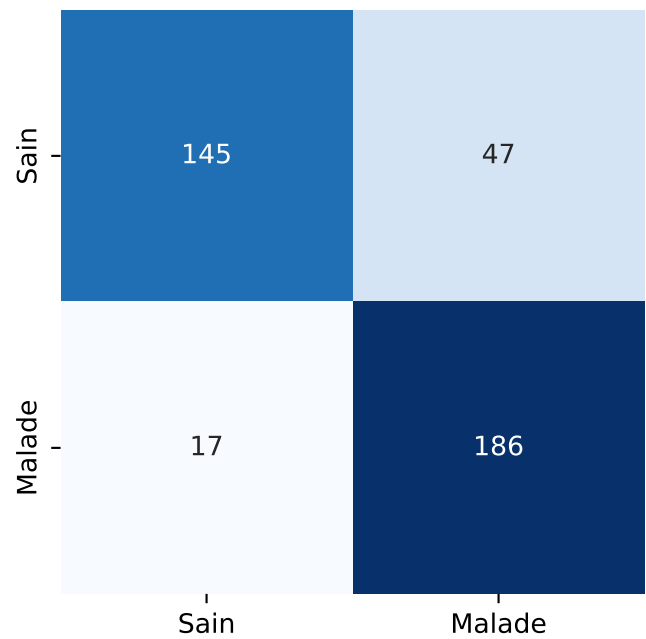
Accuracy: 83.8%

Precision: 79.8%

Recall: 91.6%

F1-Score: 85.3%

Matrice de confusion



Détail matrice

TP: 186

FP: 47

TN: 145

FN: 17

ROC (AUC = 0.915)

